

Bayesian Network Modeling of NFL Game Stats and Market Signals for Value Identification

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How the Implementation Reflects and Differs from the Proposal

The project was implemented substantially as proposed: a Naive Bayes baseline and Bayesian Network structure learning with Hill Climbing, the three-way stats / market / combined ablation, Gaussian Naive Bayes for the continuous features, and the proposed dataset of 3,990 NFL games over 15 seasons joining nflverse statistics to Sportsbook Review market data.

Three elements differ from the proposal:

- **PC algorithm — expanded.** Proposed as a secondary validation check, it became a central result: run independently, it confirmed the corrected direction of a causally impossible edge that Hill Climbing had produced, turning the dual-algorithm comparison into a contribution rather than a footnote.
- **Result framing — refined.** The proposal anticipated a binary outcome — either the network beats the baseline, or the closing line is efficient. The results supported a more specific conclusion: market signals do not improve accuracy but do improve calibration, i.e. partial efficiency.
- **Metrics — reduced.** The proposal listed Brier score, with AUC-ROC as optional. The final evaluation reports accuracy and log loss.

All other elements were delivered as proposed.